

Nithish Krishnabharathi GNANI

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WORK EXPERIENCE

Indian Institute of Science

Technical Associate

Bangalore, India

2019-2024

- Led industry-funded research projects. I was sponsored by the Center for Networked Intelligence (CNI), a Cisco CSR initiative at Robert Bosch Center for Cyber-Physical Systems (RBCCPS). (Promoted from Research Assistant to this role.)

Coca-Cola (HCCBPL)

Team Leader, Maintenance

Hyderabad, India

2017-2018

- Managed a team of 20+ associates to achieve consistent reliability of utility systems.

EDUCATION

Polytech Nice Sophia, Université Cote d'Azur

KTH Royal Institute of Technology

EIT Digital Dual Master of Science - Autonomous Systems and Intelligent Robots

Minor: Innovation and Entrepreneurship

Sophia Antipolis, France

Stockholm, Sweden

2024 - 2026

National Institute of Technology Karnataka (NITK)

Bachelor of Technology - Mechanical Engineering

Surathkal, Karnataka, India

2013 - 2017

SELECTED PROJECTS (URL for more: <https://nithishkgnani.github.io/projects/>)

BalanceBot: Simulink-Based Control Deployment on Hardware

Sep 2024 - present

- To implement various control algorithms on a two-wheeled robot to move and balance it using a model based approach.

Designing Tactile Cyber-Physical Systems

Mar 2022 - Aug 2024

- Created a testbed for real-time human-robot interaction with ultra-reliable low latency communication. Built a remote robotic arm system interfaced with a haptic device for teleoperation across continents.
- Implemented IEEE 802.1 TSN features on hardware for enhanced network reliability and latency.

5G AMMAZING

Mar 2020 - Jun 2022

- Developed a 5G mmWave system for infotainment with split data and control planes over 5G and Wi-Fi.
- Won first place (€6,500 cash award) in 5G Hackathon 2020 by Government of India among 1100+ teams.

Acoustic PUF for Sensor Security

Nov 2019 - Jul 2021

- Identified and localized sensor devices using acoustic fingerprinting for security and integrity monitoring.

Energy Harvesting IoT Sensors

Nov 2019 - Jun 2020

- Developed adaptive sampling algorithms for energy-efficient data management in IoT sensors.

Total Productive Maintenance

Nov 2017 - Apr 2018

- Improved reliability and reduced failures in Coca-Cola's bottling plant through TPM steps.

SKILLS & INTERESTS

Programming	: Python, MATLAB, Simulink, P4, C, C++, Embedded C, Shell
Hardware	: Netronome Agilio Smart NIC, Nordic nRF52 series, Arduino
Tools	: Linux, Git, ROS, Docker, LaTeX, Autodesk Fusion 360
Creative	: 3D printing, Shotcut, Snapseed
Languages	: English (IELTS band score 8/9), French (A1), Hindi, Tamil, Kannada, Sourashtra



Website

PUBLICATIONS

EdgeP4: In-Network Edge Intelligence for a Tactile Cyber-Physical System Testbed Across Cities

IEEE International Conference on Computer Communications Workshops (INFOCOM) 2024

Best Paper Award

IEEE INFOCOM '24

Towards a TSN-DetNet Intercity Testbed for Tactile Cyber-Physical Systems

IEEE International Conference on Computer Communications Workshops (INFOCOM) 2024

IEEE INFOCOM '24

Sensor Identification via Acoustic Physically Unclonable Function

Digital Threats: Research and Practice, Volume 4, Issue 2

ACM DTRAP journal

Judicious data management for sustaining an energy harvesting sensor node

Concurrency and Computation: Practice and Experience, Volume 33, Issue 23

Wiley CCPE journal